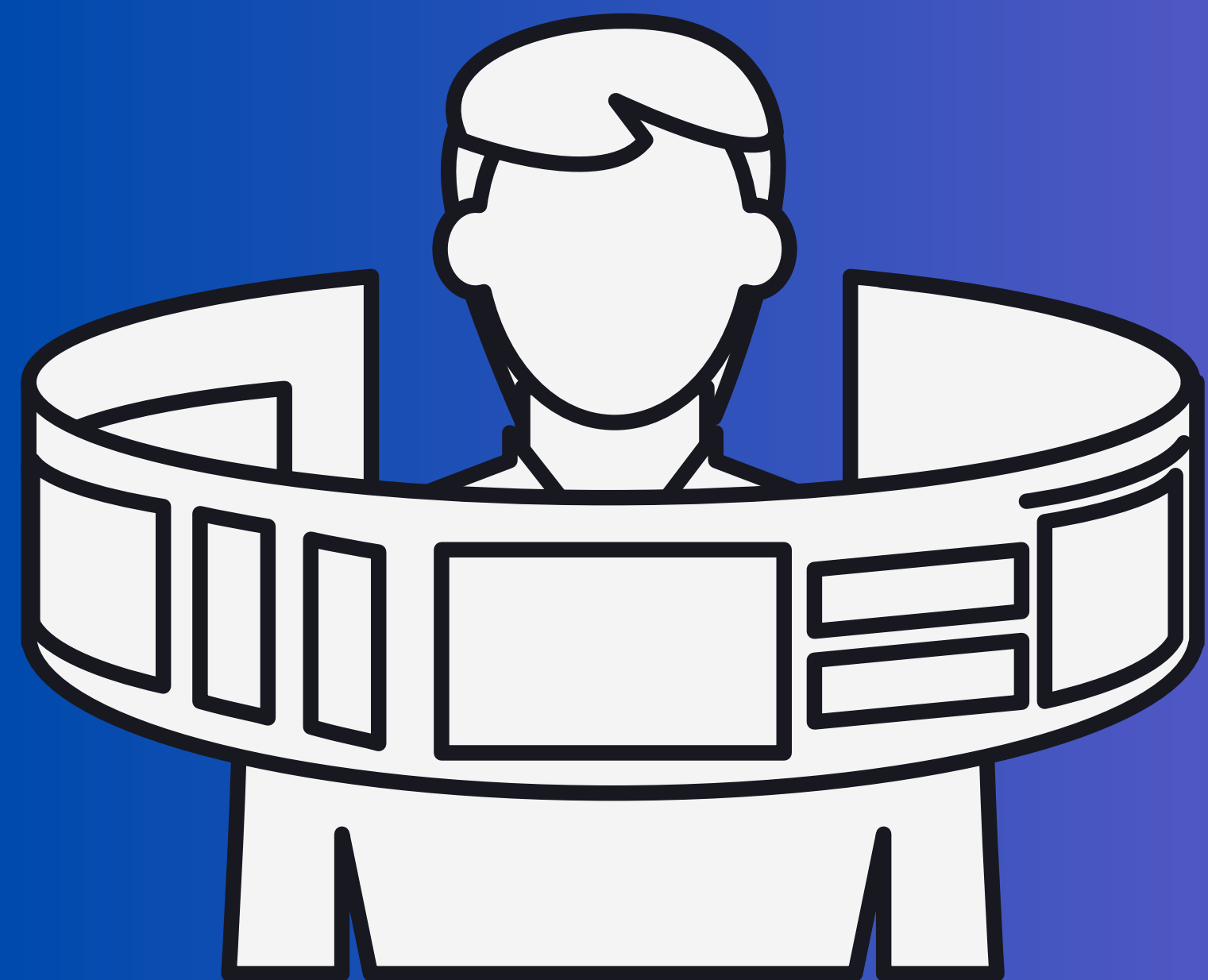




BASICS OF COMPUTER SCIENCE

WHAT IS COMPUTER SCIENCE?

WHAT IS COMPUTER SCIENCE?

- THE STUDY OF COMPUTERS AND COMPUTATIONAL SYSTEMS

SOFTWARE

- THEORY
- DESIGN
- DEVELOPMENT
- APPLICATION

WHAT IS A COMPUTER?

WHAT IS A COMPUTER?

- A MACHINE THAT CAN BE PROGRAMMED TO AUTOMATICALLY CARRY OUT SEQUENCES OF ARITHMETIC OR LOGICAL OPERATIONS (COMPUTATION)

MANAGING DATA

PROGRAMMING LANGUAGES USE **VARIABLES**

- FORMED BY A MEMORY SPACE AND AN IDENTIFIER

IDENTIFIER = VALUE

NAME OF THE VARIABLE - HOW WE REFER
TO THE VARIABLE THROUGH THE CODE

AMOUNT OF INFORMATION IN THE
MEMORY SPACE - KNOWN OR
UNKNOWN

DATA TYPES

EACH PROGRAMMING LANGUAGE RECOGNIZES AND IDENTIFIES DIFFERENT TYPES OF DATA.

BOOLEAN

- TWO POSSIBLE VALUES
 - TRUE
 - FALSE
- BOOLEAN VARIABLE HOLDS ITS STATE UNTIL IT IS CHANGED

NUMERIC

- TWO TYPES OF NUMERIC VARIABLES
 - INTEGER
 - FLOAT

STRING

- LETTERS, NUMBERS, AND SYMBOLS ARRANGED IN TEXT CHAINS
- AKA ALPHANUMERIC OR NUMERIC VARIABLES

PYTHON DATA TYPES

CATEGORY	DATA TYPE
TEXT	STR
NUMERIC	INT, FLOAT, COMPLEX
SEQUENCE	LIST, TUPLE, RANGE
MAPPING	DICT
SET	SET, FROZENSET
BOOLEAN	BOOL
BINARY	BYTES, BYTEARRAY, MEMORYVIEW
NONE	NONETYPE

SETTING THE DATA TYPE

EXAMPLE	DATA TYPE
X = "HELLO WORLD"	STR
X = 20	INT
X = 1j	COMPLEX
X = { "A", "B" }	SET
X = TRUE	BOOL
x = b"Hello"	BYTES

WHAT IS AN IDE?

WHAT IS AN IDE?

- INTEGRATED DEVELOPMENT ENVIRONMENT
- SOFTWARE APPLICATION THAT HELPS DEVELOP SOFTWARE CODE EFFICIENTLY
- CONSISTS OF
 - A SOURCE-CODE EDITOR
 - BUILD AUTOMATION TOOLS
 - DEBUGGER

JETBRAINS IDE



JETBRAINS IDEs

Make it happen.
With code.

Enjoy an exceptional developer experience.

Choose your IDE

JETBRAINS IDE



IntelliJ IDEA

IDE for Java and Kotlin developers

Java Kotlin Spring



PyCharm

IDE for Python developers and data scientists

Python Django Jupyter



PhpStorm

IDE for PHP developers

PHP Laravel Symfony



GoLand

IDE for Go developers

Go (Golang) JavaScript TypeScript



Rider

IDE for .NET and game developers

C# .NET ASP.NET



CLion

IDE for C and C++ developers

C C++ CMake



RustRover

IDE for Rust developers

Rust SQL JavaScript



WebStorm

IDE for JavaScript and TypeScript developers

JavaScript TypeScript React



RubyMine

IDE for Ruby and Rails developers

Ruby on Rails (RoR) Hotwire RuboCop



DataGrip

Tool for multiple databases

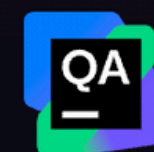
Databases SQL NoSQL



ReSharper

Extension that makes Visual Studio a much better IDE

C# .NET ASP.NET



Aqua

IDE for test automation engineers

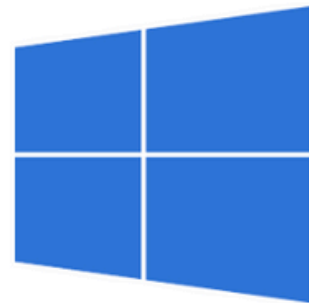
Selenium Selenide Playwright

VISUAL STUDIO CODE DOWNLOAD

[HTTPS://CODE.VISUALSTUDIO.COM/DOWNLOAD](https://code.visualstudio.com/download)

Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



↓ Windows

Windows 10, 11

User Installer [x64](#) [Arm64](#)
System
Installer [x64](#) [Arm64](#)
.zip [x64](#) [Arm64](#)
CLI [x64](#) [Arm64](#)



↓ .deb

Debian, Ubuntu

↓ .rpm

Red Hat, Fedora, SUSE

.deb [x64](#) [Arm32](#) [Arm64](#)
.rpm [x64](#) [Arm32](#) [Arm64](#)
.tar.gz [x64](#) [Arm32](#) [Arm64](#)
Snap [Snap Store](#)
CLI [x64](#) [Arm32](#) [Arm64](#)



↓ Mac

macOS 10.15+

.zip [Intel chip](#) [Apple silicon](#) [Universal](#)
CLI [Intel chip](#) [Apple silicon](#)

INSTALLING EXTENSIONS

EXTENSIONS: MARKETPLACE

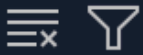


Release Notes: 1.96.0

Extension: Python



Python



Python

Python language support with extension access points for IntelliSense (Pylance), Debugg...

Microsoft



Python

Microsoft | 153,142,465 | (60)

Python language support with extension access points for IntelliSe...

Disable Uninstall Switch to Pre-Release Version Auto Update

DETAILS FEATURES CHANGELOG EXTENSION PACK

Python extension for Visual Studio Code

A Visual Studio Code extension with rich support for the Python language (for all actively supported Python versions), providing access points for extensions to seamlessly integrate and offer support for IntelliSense (Pylance), debugging (Python Debugger), formatting, linting, code navigation, refactoring, variable explorer, test explorer, and more!

Support for [vscode.dev](#)

The Python extension does offer some support when running on [vscode.dev](#) (which includes [github.dev](#)). This includes partial IntelliSense for open files in the editor.

Installed extensions

The Python extension will automatically install the following extensions by default to provide the best Python development experience in VS Code:

- Pylance - to provide performant Python language support



Installation

Identifier	ms-python.python
Version	2024.22.2
Last Updated	2025-01-22, 12:14:44
Cache	848 B

Marketplace

Published	2016-01-19, 09:03:11
Last Released	2025-01-24, 04:42:55

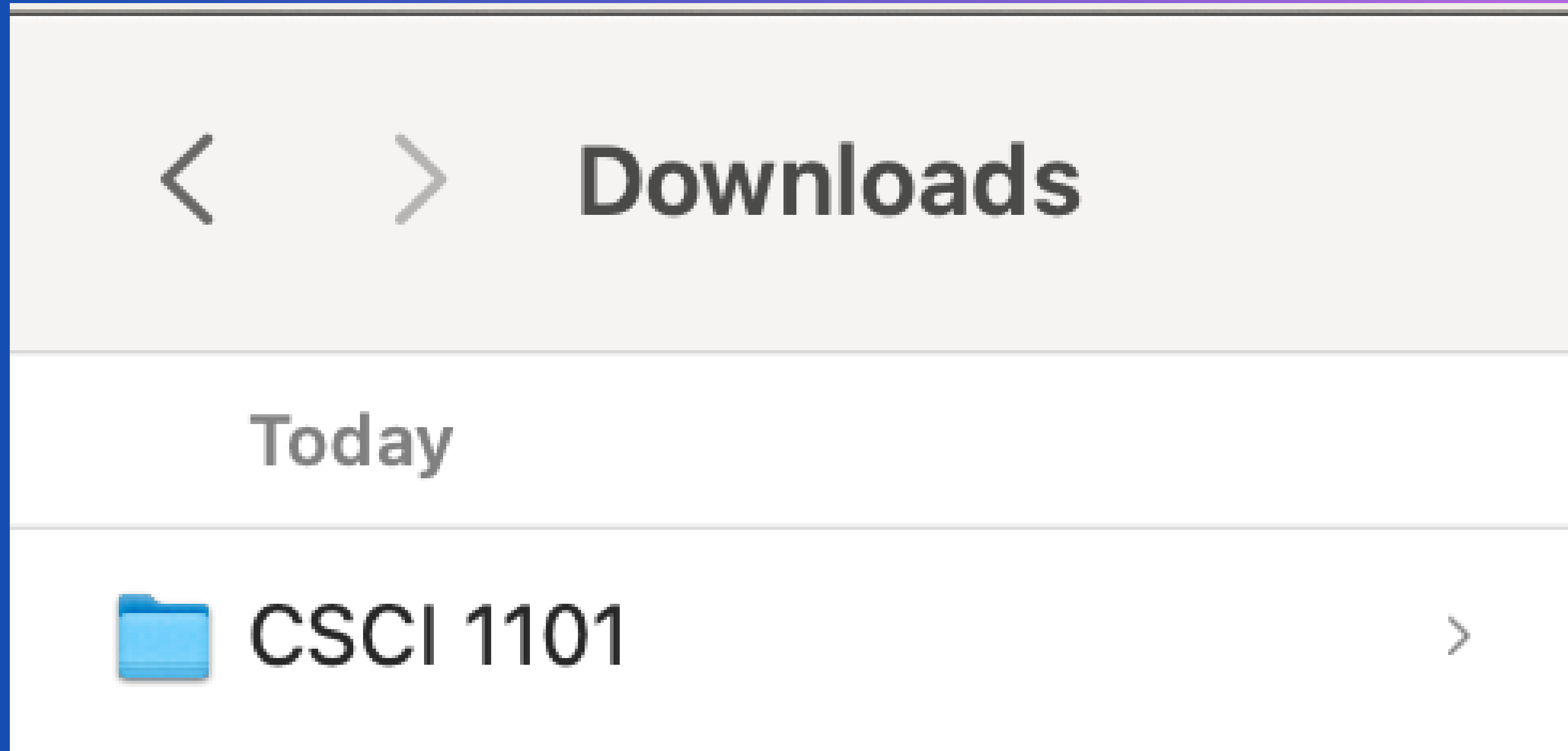
Categories

- Programming Languages
- DebuggersOther
- Data Science
- Machine Learning

BEFORE RUNNING PYTHON SCRIPT

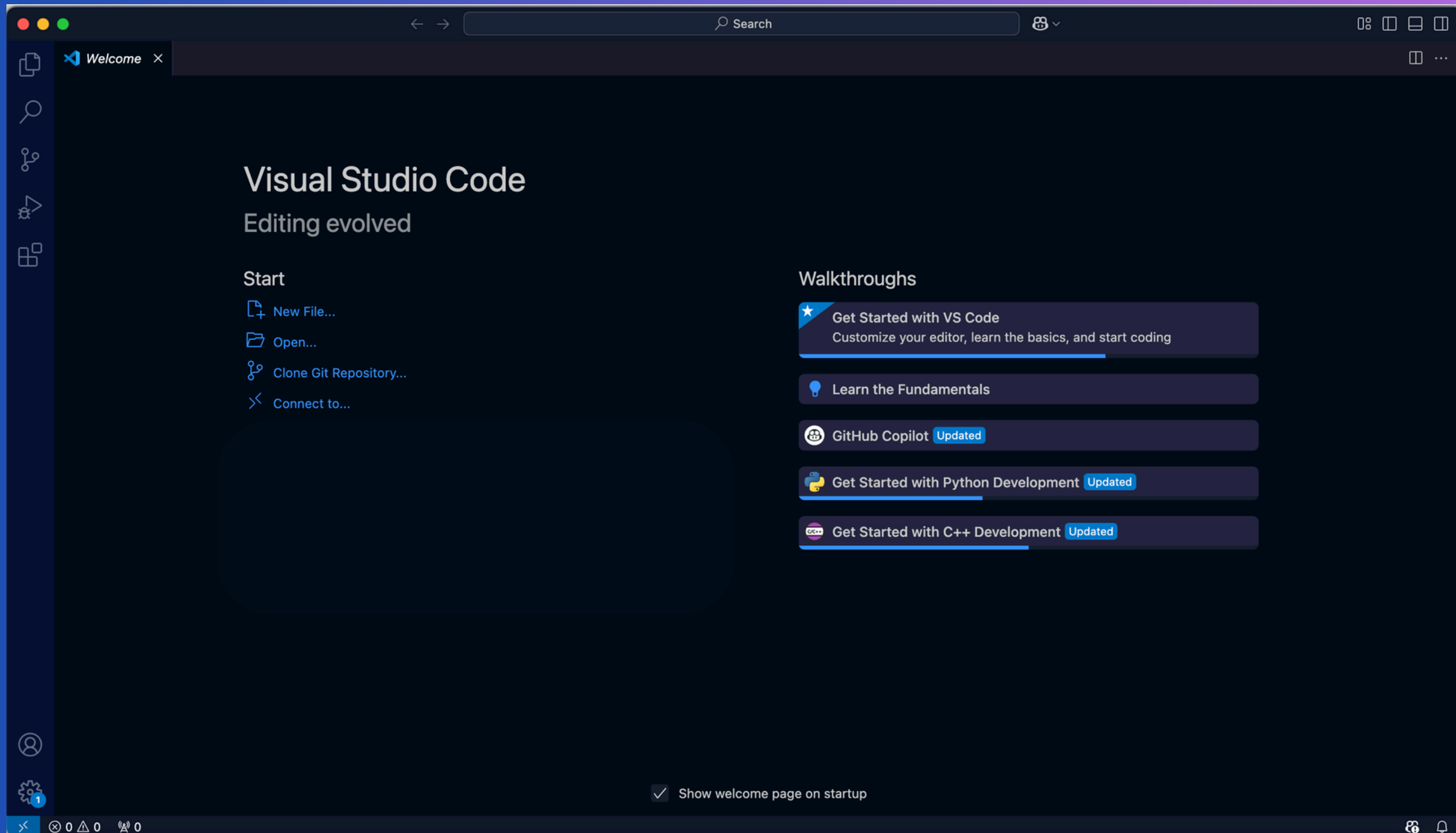
BEFORE RUNNING PYTHON SCRIPT

- CREATE A FOLDER AND TITLE IT



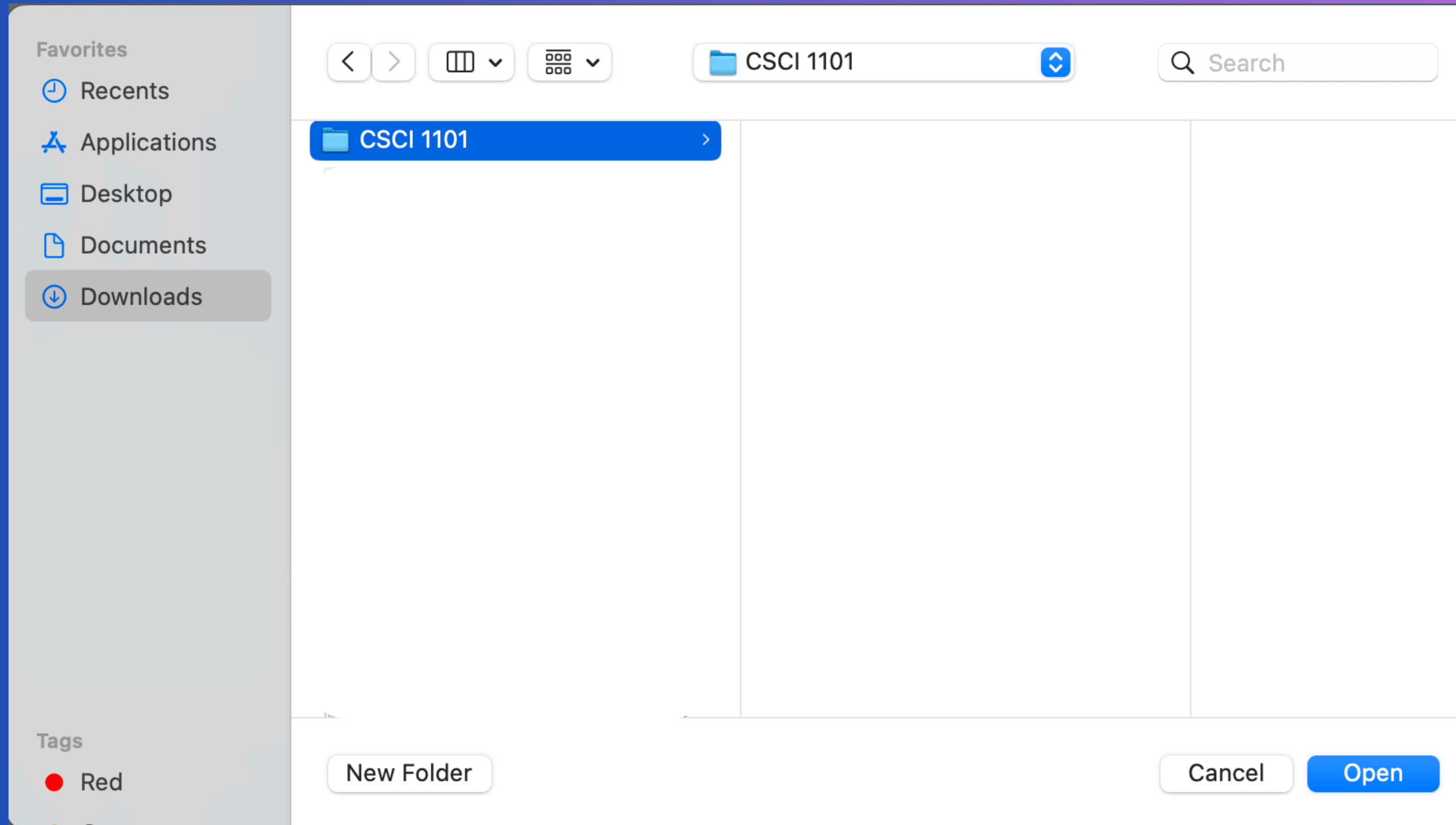
BEFORE RUNNING PYTHON SCRIPT

- GO TO VISUAL STUDIO CODE



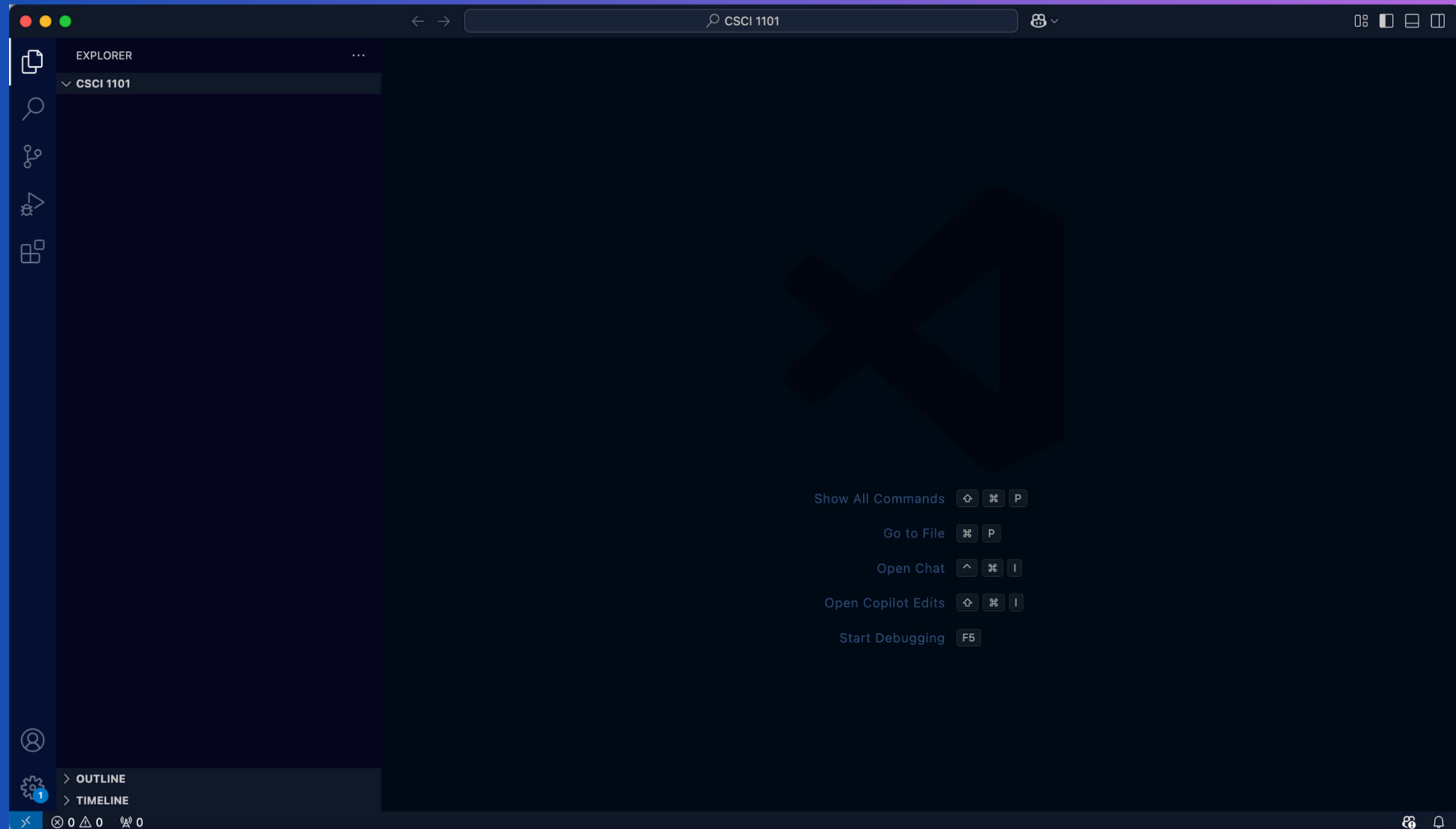
BEFORE RUNNING PYTHON SCRIPT

- CLICK OPEN AND CHOOSE FOLDER



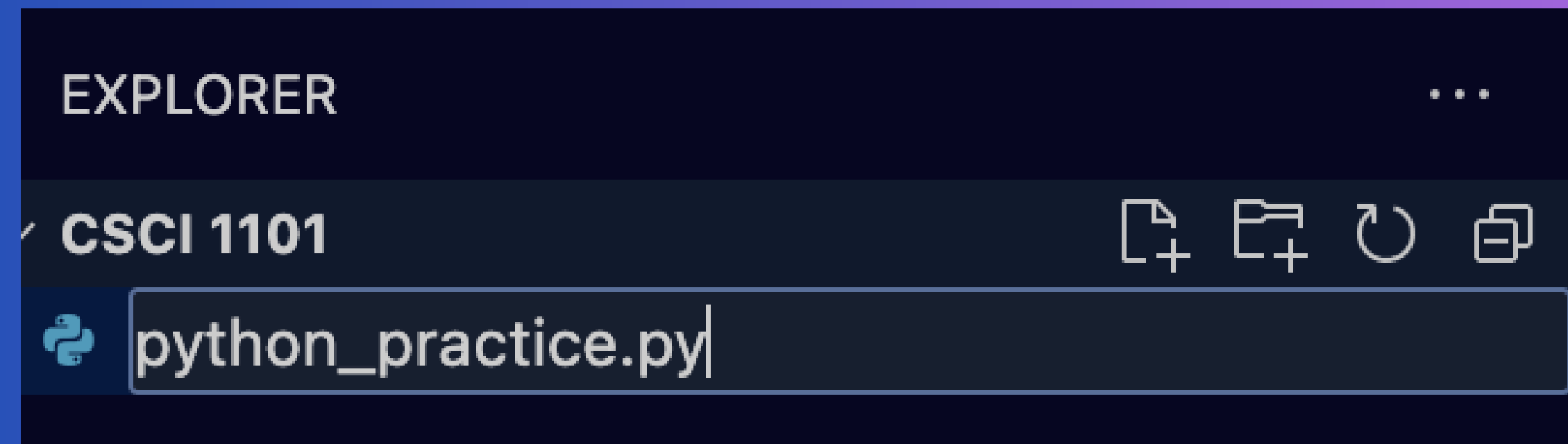
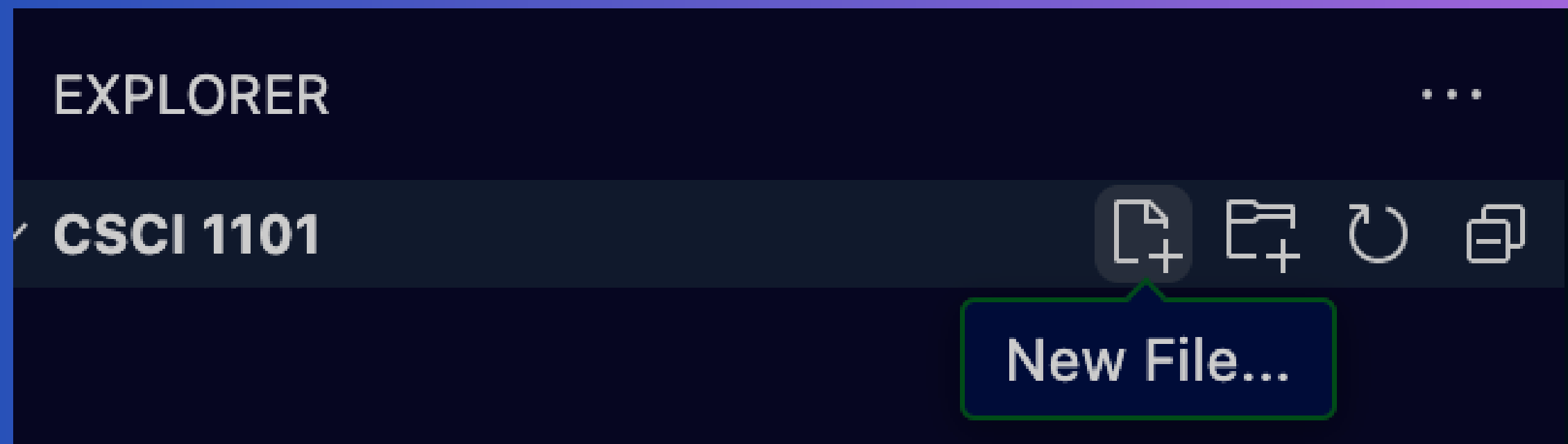
BEFORE RUNNING PYTHON SCRIPT

*WHAT VISUAL STUDIO CODE LOOKS LIKE



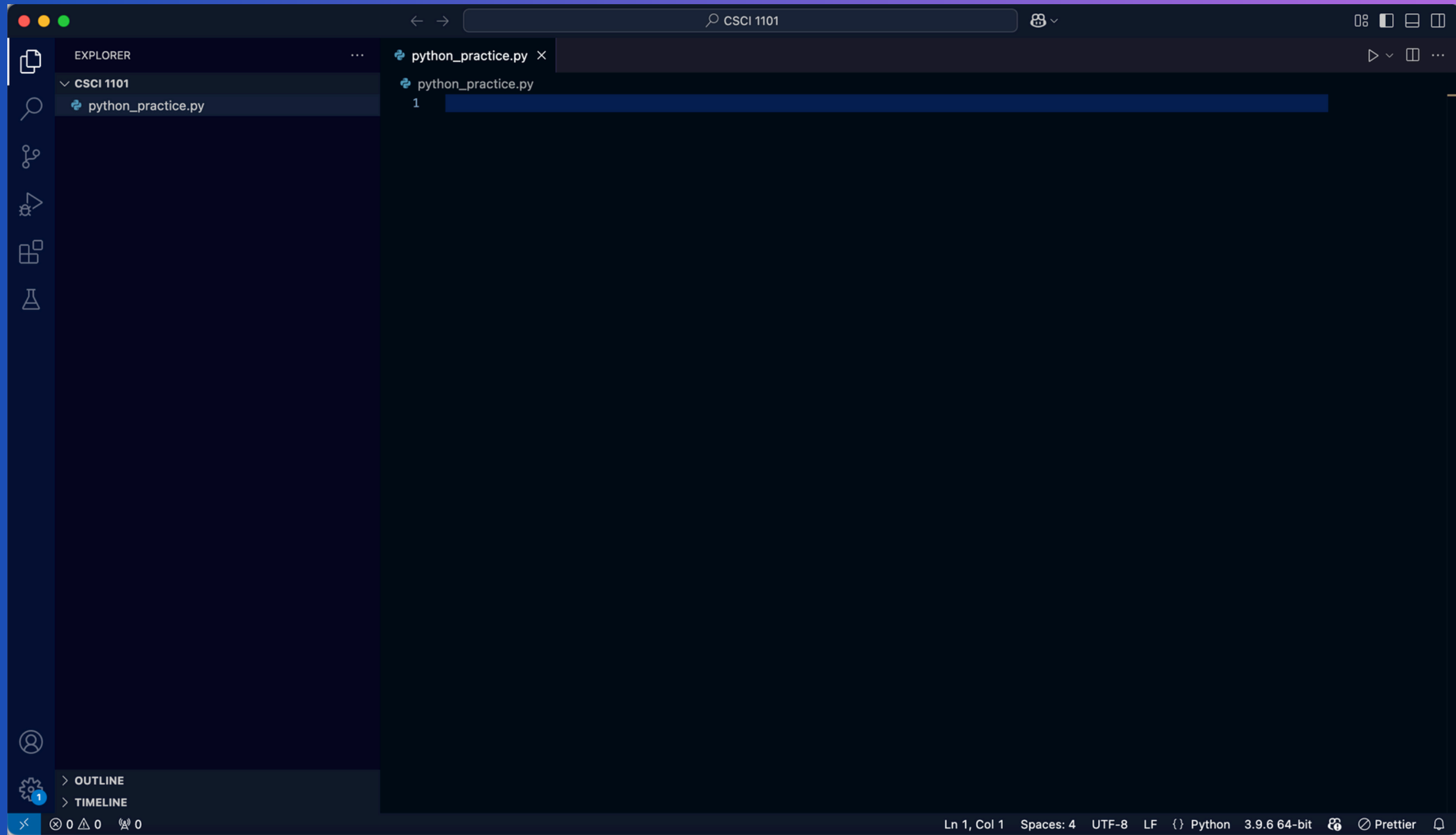
BEFORE RUNNING PYTHON SCRIPT

- CREATE A NEW FILE INSIDE THE FOLDER.
- INCLUDE THE .PY AT THE END OF THE TITLE



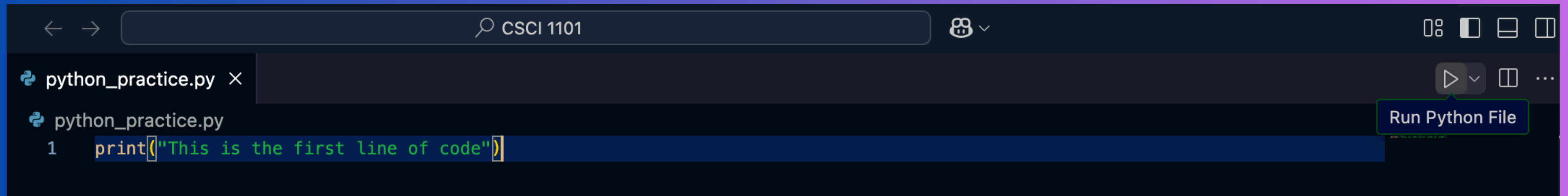
BEFORE RUNNING PYTHON SCRIPT

*SHOULD LOOK LIKE THIS

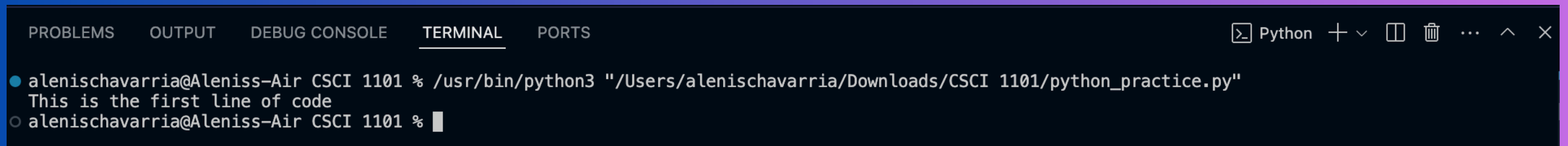


BEFORE RUNNING PYTHON SCRIPT

- WRITE PYTHONIC CODE
- RUN IT BY CLICKING THE 'PLAY'-LIKE BUTTON



*AFTER RUNNING CODE



CAREERS IN COMPUTER SCIENCE

GAME DEVELOPMENT

GAME PROGRAMMER/DEVELOPER

- DESIGNS AND IMPLEMENTS THE CORE GAMEPLAY MECHANICS, SUCH AS CHARACTER MOVEMENT, COMBAT SYSTEMS, AI BEHAVIOR, AND MORE. THEY COLLABORATE WITH GAME DESIGNERS AND ARTISTS TO BRING THE GAME CONCEPT TO LIFE.
- C++, C#, PYTHON, JAVA, JAVASCRIPT
- \$50,000 - \$75,000 PER YEAR *

GAME ENGINEER PROGRAMMER

- WORKS ON THE DEVELOPMENT AND OPTIMIZATION OF GAME ENGINES (E.G., UNREAL, UNITY, CUSTOM ENGINES) TO ENSURE THE FRAMEWORK RUNS SMOOTHLY AND SUPPORTS THE DEVELOPMENT TEAM'S NEEDS
- C++, C#, PYTHON
- \$70,000 - \$90,000 PER YEAR *

SOFTWARE ENGINEERING

FRONT-END DEVELOPER

- FOCUSES ON THE CLIENT-SIDE OF APPLICATIONS, BUILDING THE USER INTERFACE (UI) AND ENSURING AN ENGAGING USER EXPERIENCE (UX). THEY WORK ON LAYOUTS, INTERACTIVITY, AND RESPONSIVENESS
- [FOR WEB APPS] HTML, CSS, JAVASCRIPT, TYPESCRIPT
- REACT, ANGULAR, VUE.JS
- \$60,000 - \$80,000 PER YEAR *

DEVOPS PROGRAMMER

- BRIDGES DEVELOPMENT AND OPERATIONS BY AUTOMATING, MONITORING, AND MANAGING DEPLOYMENT PIPELINES, INFRASTRUCTURE, AND PERFORMANCE MONITORING
- PYTHON, GO, RUBY
- (TOOLS) DOCKER, KUBERNETES, JENKINS
- \$75,000 - \$95,000 PER YEAR *

*all pay is estimated entry-level pay

WEB DEVELOPMENT

WEB DESIGNER

- SPECIALIZES IN CREATING THE VISUAL LAYOUT, AESTHETICS, AND USABILITY OF WEBSITES. WEB DESIGNERS FOCUS ON DESIGN PRINCIPLES, UX/UI, AND OFTEN COLLABORATE WITH DEVELOPERS FOR IMPLEMENTATION
- HTML, CSS, JAVASCRIPT
- (TOOLS) FIGMA. ADOBE XD, SKETCH
- \$50,000 - \$65,000 PER YEAR *

WEB PERFORMANCE ENGINEER

- OPTIMIZES WEBSITE PERFORMANCE BY REDUCING LOAD TIMES, IMPROVING ACCESSIBILITY, AND ENSURING RESPONSIVENESS ACROSS ALL DEVICES
- HTML, CSS, JAVASCRIPT
- (TOOLS) FIGMA. ADOBE XD, SKETCH
- \$60,000 - \$80,000 PER YEAR *

*all pay is estimated entry-level pay

CYBERSECURITY

PENETRATION TESTER(ETHICAL HACKER)

- SPECIALIZES IN CREATING THE VISUAL LAYOUT, AESTHETICS, AND USABILITY OF WEBSITES. WEB DESIGNERS FOCUS ON DESIGN PRINCIPLES, UX/UI, AND OFTEN COLLABORATE WITH DEVELOPERS FOR IMPLEMENTATION
- PYTHON, RUBY, JAVASCRIPT, C/C++, BASH/SHELL SCRIPTING
- \$70,000 - \$90,000 PER YEAR *

CRYPTOGRAPHER

- DEVELOPS AND IMPLEMENTS ENCRYPTION ALGORITHMS TO SECURE SENSITIVE DATA. THEY ALSO ANALYZE AND BREAK CRYPTOGRAPHIC SYSTEMS TO ENSURE THEIR ROBUSTNESS
- PYTHON, C, C++, JAVA
- \$80,000 - \$100,000 PER YEAR *

*all pay is estimated entry-level pay

ARTIFICIAL INTELLIGENCE

NLP ENGINEER (NATURAL LANGUAGE PROCESSING)

- DEVELOPS APPLICATIONS TO PROCESS AND ANALYZE HUMAN LANGUAGE, SUCH AS CHATBOTS, VOICE ASSISTANTS, SENTIMENT ANALYSIS TOOLS, AND TEXT SUMMARIZERS
- PYTHON, JAVA, R, SQL
- \$80,000 - \$110,000 PER YEAR *

AI RESEARCH SCIENTIST

- FOCUSES ON ADVANCING THE FIELD OF ARTIFICIAL INTELLIGENCE BY DEVELOPING NEW ALGORITHMS, ARCHITECTURES, AND TECHNOLOGIES.
- PYTHON, C++. JAVA, MATLAB
- \$90,000 - \$120,000 PER YEAR *

*all pay is estimated entry-level pay